

Nilabhra Patra

nlbhrptr@hotmail.com ❖ (+49) 15218044701 ❖ <https://www.linkedin.com/in/nilbro/>

WORK EXPERIENCE

ProSiebenSat.1 Media SE

June 2025 – Present

Software Engineer (AI/Recommendation Systems)

Munich, Germany

- **Involved in recommendation serving implementation for Joyn streaming services via REST API and GraphQL with Kafka event triggers**, collaborating with cross-functional stakeholders to define integration patterns.
- **Architected and implemented phased migration of real-time recommendation system from monolithic GCE instance to decoupled serverless services**, reducing deployment complexity, enabling independent service scaling, and improving cost efficiency by decoupling Kafka stream processing from inference serving.
- **Established a unified A/B testing and feature-flagging framework using Growthbook** across multiple teams, aiming to standardise experimentation infrastructure company-wide.
- **Contributed to model training pipelines using Vertex AI (Kubeflow)**, supporting scalable ML operations on GCP infrastructure.
- **Drove technical and AI excellence initiatives**: Optimised Docker build pipeline and implemented semantic release automation. Modernised observability stack with Datadog and Teams integration. Developed and used Claude Skills for agentic coding, GitOps, and systematic debugging.

ProSiebenSat.1 Media SE

May 2023 – June 2025

Software Engineer (AI/Cloud/Data)

Munich, Germany

- **Reduced ETL runtime by 70%**, optimising PostgreSQL queries, tuning RDS database resources, and orchestrating workflows with Glue and AWS Step to cut execution time from 5 to 1.5 hours.
- **Accelerated ML development** by implementing Metaflow, enabling scalable, reproducible Data pipelines and ML training workflows on AWS and Databricks.
- **Extended and optimised existing RESTful APIs** using FastAPI and SQLAlchemy, ensuring backwards compatibility while significantly improving performance and robustness for large-scale data queries.
- **Automated end-to-end testing and deployment** via CI pipelines, boosting delivery speed and reliability through unit/integration testing and cloud deployment.
- **Introduced and championed Terraform and uv**, standardising infrastructure-as-code and Python dependency management, leading to faster environment provisioning and consistent development workflows.
- **Led cross-functional initiatives** to build internal libraries, Terraform modules, and CI components, improving reusability and flexibility within the department.

RTL Deutschland

March 2021 – April 2023

Data Engineer (Knowledge Engineering)

Cologne, Germany

- **Played a key role** in launching a new data product from scratch, contributing to early-stage **design, architecture, and engineering decisions**.
- **Designed and evolved** our media ontology using Protégé, RDF, SPARQL, and semantic technologies, ensuring **consistency and extensibility**.
- **Built a Kubernetes-native data pipeline** on GCP to enable **scalable, cloud-native processing** with BigQuery and PubSub.
- **Maintained scalable and reliable graph databases** on GKE, supporting **complex semantic queries** and large-scale data relationships.

Robert Bosch GmbH

Internship (Data Engineering)

March 2020 – June 2020

Stuttgart, Germany

- **Designed and implemented real-time ETL pipelines** for an Industry 4.0 use case, using **MongoDB, Kafka and Spark Streaming**.
- **Developed a real-time monitoring dashboard using InfluxDB and Grafana** to visualise streaming sensor data and operational metrics.
- **Self-hosted and managed the complete infrastructure stack**, handling deployment, configuration, and maintenance of all pipeline components.

MASTER'S THESIS

Robert Bosch GmbH

Master's Thesis

June 2020 – March 2021

Stuttgart, Germany

- **Conducted domain analysis of assembly line processes for industrial pumps**, collaborating with manufacturing experts to understand operational workflows and data requirements.
- **Extended existing manufacturing ontology using OWL and Protégé**, modelling domain-specific concepts and relationships to support semantic data integration.
- **Integrated heterogeneous data sources (XML, CSV, Excel) into the ontology framework**, enabling unified semantic queries via SPARQL across previously siloed manufacturing data.

EDUCATION

Fachhochschule Dortmund

M.Eng Embedded Systems for Mechatronics

Oct. 2018 – April 2021

TECHNICAL SKILLS

- **Programming & Scripting:** Python, SQL, SPARQL, Bash
- **Cloud Platforms:** GCP, AWS
- **DevOps & Infrastructure:** Git, Docker, Kubernetes (GKE), GitLab CI, Terraform, MLflow
- **Testing & Automation:** PyTest, Teams Workflows
- **Machine Learning & Data Science:** CatBoost, SHAP, Optuna, Random Forest, Neural Networks (FNN, RNN), KNN, Feature Engineering, Nvidia Triton, JAXlib
- **Data Engineering:** pandas, NumPy, matplotlib, seaborn, Snowflake, Exploratory Data Analysis (EDA)
- **Backend:** FastAPI, Pydantic, Ariadne, SlowAPI, Kafka, Faststream, Redis, Click
- **Development Tools:** Claude Code, GitHub CoPilot

CERTIFICATIONS, SKILLS AND INTERESTS

- **Certifications:** [AWS Certified Solutions Architect - Associate](#)
- **Languages:** English (C2), German (B2)